

$S = (U, V, R)$

- primitive event $\rightarrow (X=x), X \in V \ \& \ x \in R(X)$

- causal formula over $S \rightarrow$

$[Y_1 \leftarrow y_1, \dots, Y_K \leftarrow y_K] \varphi$

① φ is a boolean combination of primitive events

② $y_1, \dots, y_K$ are distinct variables in V

③ $x_i \in R(Y_i)$

• A primitive event is a claim that a specific internal variable x has taken on a specific value x from its allowed Range $R(X)$

• Boolean combination φ represents a more complex statement built out of primitive events & logical operators ($\wedge, \vee, !$)

ex- $(X_1=1) \wedge (X_2=0)$

Example 3.4
pg 8, Sander
Beckers
Abstracting
causal
models

$i = \{1, 2\}$ for M_i , M_i is deterministic causal

model, signature (U_i, V_i, R_i) & $\vec{u}_i$ is a fixed context in M_i ; let $\vec{v}_i \in R_i(V_i)$ s.t. $(M_i, \vec{u}_i) \models V_i = \vec{v}_i$.

I_i is the empty intervention;

Pr_i puts probability 1 on $\vec{u}_i$

T_i maps all elements $R(V_i) \rightarrow \vec{v}_{3-i}$

w_i is the identity map from $I_i \rightarrow I_{3-i}$

$\Rightarrow (M_i, Pr_i)$ is an exact $(T_i - w_i)$ transformation of (M_{3-i}, Pr_{3-i})

$$\left. \begin{array}{l} T_1: R(V_1) \rightarrow \vec{v}_2 \\ w_1: I_1 \rightarrow I_2 \end{array} \right\}$$

$$\tau(M_L(\underbrace{\vec{u}_L}_{\text{Low Level context}}, \underbrace{\vec{Y} \leftarrow \vec{y}}_{\text{applying the intervention in low level model}})) = M_H(\underbrace{\tau'(\vec{u}_L), w(\vec{Y} \leftarrow \vec{y})}_{\text{abstracting the noise \& intervention first}})$$

Low level model
→ Abstracting using the abstraction function

abstracting the noise & intervention first

→ Applying directly on high level abstracted model